Math Inquirers

Extension Workbook

with answers

Level 4

Self-study | Ages 9-11



My name _____

About this book

Math Inquirers Extension Workbook, Level 4
Self-study practice with answers.

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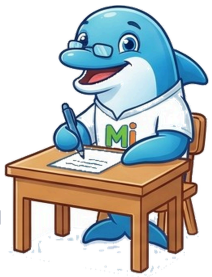
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Edition 1, 2026. First printing.

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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Factor hunters	Not in this proof edition				☐
2	Big number puzzles	Not in this proof edition				☐
3	Operation Detectives	6-7	8-9	10 and 12	11	☐
4	Fraction frontier	Not in this proof edition				☐
5	Decimal detectives	Not in this proof edition				☐
6	Angle turns	Not in this proof edition				☐
7	Area architects II	Not in this proof edition				☐
8	Symmetry and nets	Not in this proof edition				☐
9	Data detectives	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

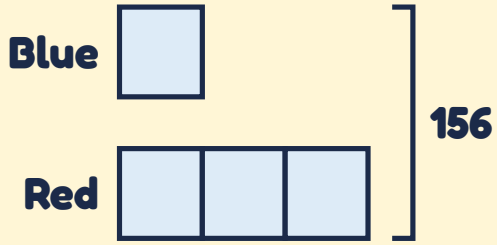
Learn it: Four operations

My goal. I can solve multi-step problems with the four operations. I can use bar models for "times as many" and work backwards through a number machine.

A bar model shows a "times as many" story. Each equal part is one unit. To go back to the start of a number machine, undo each step in the opposite order.

Worked example 1

Mr Tan has 3 times as many red pens as blue pens. He has 156 pens in all. How many blue pens does he have?

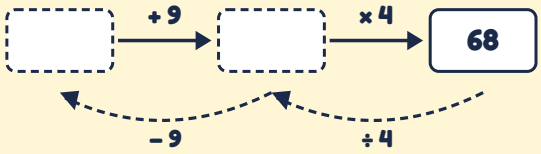


1. Blue is 1 unit and red is 3 units. That is 4 units.
2. 4 units = 156, so 1 unit = 156 ÷ 4 = 39.

Answer: He has 39 blue pens.

Worked example 2

I think of a number. I add 9 and then multiply by 4. The answer is 68. What is my number?



1. Undo the steps in the opposite order.
2. Undo × 4 with ÷ 4: 68 ÷ 4 = 17. Undo + 9 with − 9: 17 − 9 = 8.

Answer: My number is 8. Check: 8 + 9 = 17 and 17 × 4 = 68.

How to solve it

1. Bar models: draw one bar for each quantity. Count the units.
2. Divide the total by the number of units.
3. Machines: start from the end and undo each step. Check by going forward.

Opposites

- + and − undo each other.
- × and ÷ undo each other.
- "4 times as many" is 4 units.

Your turn

Answer at the back of the book.

Try A shop has 5 times as many pens as rulers. There are 114 pens and rulers. How many rulers are there? ☐

Start here

Do these first. Take your time.

1 **Work these out.**

(a) 3706 × 4 = ☐

(b) 85 × 47 = ☐

(c) 7524 ÷ 6 = ☐

(d) 8150 ÷ 5 = ☐

Working: _____

_____

2 **Round each number to the nearest 1000. Then estimate the answer.**

(a) 4986 × 4 is about ☐

(b) 9120 ÷ 3 is about ☐

(c) 7020 – 2980 is about ☐

3 **A hall has 28 rows of 36 chairs. Then 45 chairs are broken and taken away. How many chairs are left in the hall?** ☐ **chairs**

Working: _____

_____

4 **A baker packs 5468 buns into boxes of 8.**

(a) How many full boxes does she fill? ☐ boxes

(b) How many buns are left over? ☐ buns

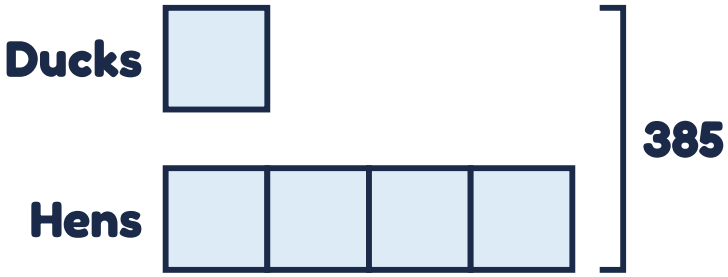
Working: _____

Start here

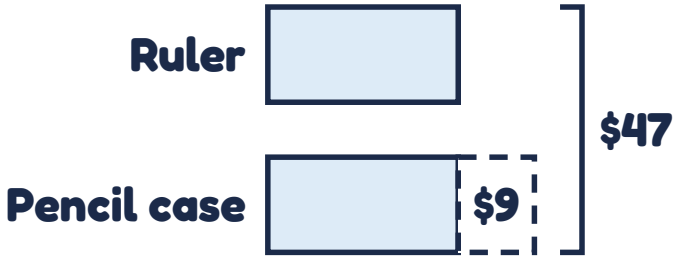
Keep going. Check your work.

5 A farm has 4 times as many hens as ducks. There are 385 hens and ducks in all.

How many ducks are there? ☐ ducks



6 A ruler and a pencil case cost $47 together. The pencil case costs $9 more than the ruler.



(a) The ruler costs $ ☐ (b) The pencil case costs $ ☐

7 I think of a number. I multiply it by 6, take away 9, and then divide by 3. The answer is 31. Find my number.



My number is ☐

Start here

Nearly there. You can do it.

9 **Find the missing numbers.**

(a) 7 × ☐ = 4249

(b) ☐ ÷ 9 = 124

(c) 5000 − ☐ = 2738

(d) ☐ + 1905 = 6400

10 **Mrs Lee buys 6 notebooks at $3 each and 4 pens at $2 each. She pays with a $50 note. How much change does she get? $** ☐

Working: _____

_____

11 **Each number after the first two is the sum of the 2 numbers before it. Fill in the missing numbers.**

(a) 4, 9, ☐, 22, ☐

(b) ☐, 6, 11, ☐, 28

12 **Kumar picked 3 times as many mangoes as Aisha. They picked 480 mangoes in all. How many more mangoes did Kumar pick than Aisha?** ☐ **mangoes**

Draw a bar model here.

_____

13 **Siti reads 24 pages a day for 15 days. Ali reads 18 pages a day for 21 days.**

(a) Who reads more pages? Siti Ali

(b) How many more? ☐ pages

Working: _____

_____

Think harder

Hints are upside down. Try first!

1 **Beyond Level 4: Order of operations**
Rule: do brackets first, then × and ÷, then + and –. Work from the left.

(a) 60 – 4 × 9 + 12 ÷ 3 = ☐

(b) 5 × (13 + 7) – 6 = ☐

Working: _____

_____

Hint. In part (a), do 4 × 9 and 12 ÷ 3 before anything else.

2 **Each number after the first two is the sum of the 2 numbers before it. The 5th number is 41 and the 6th number is 66. What is the 1st number?** ☐

Working: _____

_____

_____

Hint. Work backwards. The 4th number is 66 – 41.

3 **Mei has 5 times as many stickers as Lily. Mei gives 36 stickers to Lily. Then they have the same number of stickers.**

(a) How many stickers does Lily have at first? ☐

(b) How many stickers does Mei have at first? ☐

Draw a bar model here.

_____

_____

Hint. After the gift, Mei has 36 + 36 more than 1 unit.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 4: Signs and brackets

Brackets and the order of operations are on the Think harder page.

(a) Write + – × or ÷ in each circle: 9 ○ 4 ○ 6 = 30

(b) Put ONE pair of brackets in 7 + 3 × 8 – 4 to make 76. Write the numbers inside the brackets: ☐ + ☐

Try: _____

_____

Hint. For (a), estimate first. A big answer needs ×.

2 Beyond Level 4: Order of operations

The same number goes in all three boxes. What is the number?

☐ × ☐ + ☐ = 56

☐

Try: _____

_____

Hint. Try 6, then 7, then 8. Is the answer too small or too big?

3 Ben and Mei both save for a bike. Plan A: start with $15 and save $8 each week. Plan B: start with $0 and save $13 each week.

Week	0	1	2	3	4
Plan A	$15				
Plan B	$0				

(a) In which week do both plans have the same amount? Week ☐

(b) How much is that? $ ☐

(c) After 10 weeks, Plan B has $ ☐ more than Plan A.

Hint. Fill in the table. Look at the gap between the plans each week.

Check yourself

No hints. Try each one on your own.

1 **Work out.** **4728 × 5 =** ☐ **9135 ÷ 7 =** ☐

Working: _____

2 **Ali saves $15 a week for 12 weeks. Then he spends $64. How much money does he have left? $** ☐

Working: _____

3 **There are 4 times as many red cars as blue cars. There are 275 cars in all. How many red cars are there?** ☐

Working: _____

4 **I think of a number. I add 17, multiply by 3 and take away 20. The answer is 94. What is my number?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Factor hunters	☐	☐	☐	☐	/ 4	
2	Big number puzzles	☐	☐	☐	☐	/ 4	
3	Operation Detectives	☐	☐	☐	☐	/ 4	
4	Fraction frontier	☐	☐	☐	☐	/ 4	
5	Decimal detectives	☐	☐	☐	☐	/ 4	
6	Angle turns	☐	☐	☐	☐	/ 4	
7	Area architects II	☐	☐	☐	☐	/ 4	
8	Symmetry and nets	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 4.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 3 · Operation Detectives

Your turn
page 6

Answer: 19
Why: Rulers are 1 unit and pens are 5 units: 6 units in all. 6 units = 114, so 1 unit = 114 ÷ 6 = 19 rulers.

START HERE

1
page 7

Answer: (a) 14 824 (b) 3995 (c) 1254 (d) 1630
Why: (a) 3706 × 4 = 12 000 + 2800 + 24 = 14 824. (b) 85 × 47 = 85 × 40 + 85 × 7 = 3400 + 595 = 3995. (c) 6 × 1254 = 7524. (d) 5 × 1630 = 8150.

2
page 7

Answer: (a) 20 000 (b) 3000 (c) 4000
Why: (a) 4986 rounds to 5000 and 5000 × 4 = 20 000. (b) 9120 rounds to 9000 and 9000 ÷ 3 = 3000. (c) 7020 rounds to 7000 and 2980 rounds to 3000, and 7000 − 3000 = 4000.

3
page 7

Answer: 963
Why: 28 × 36 = 1008 chairs at first. 1008 − 45 = 963 chairs are left.

4
page 7

Answer: (a) 683 (b) 4
Why: 5468 ÷ 8 = 683 with a remainder of 4, because 8 × 683 = 5464 and 5468 − 5464 = 4.

5
page 8

Answer: 77
Why: Ducks are 1 unit and hens are 4 units: 5 units in all. 5 units = 385, so 1 unit = 385 ÷ 5 = 77 ducks.

6
page 8

Answer: (a) 19 (b) 28
Why: Take away the extra $9: $47 − $9 = $38. Two equal parts are $38, so the ruler costs $38 ÷ 2 = $19. The pencil case costs $19 + $9 = $28.

7
page 8

Answer: 17
Why: Work backwards. 31 × 3 = 93. 93 + 9 = 102. 102 ÷ 6 = 17. Check: 17 × 6 = 102, 102 − 9 = 93, 93 ÷ 3 = 31.

8
page 8

Answer: (a) 4 (b) 4 (c) 4
Why: 6 × 3 = 18, 18 + 12 = 30, 30 ÷ 3 = 10, 10 − 6 = 4. 15 × 3 = 45, 45 + 12 = 57, 57 ÷ 3 = 19, 19 − 15 = 4. Multiplying by 3 and then dividing by 3 gives back the number. The 12 ÷ 3 adds 4, and taking away the number leaves 4.

9
page 9

Answer: (a) 607 (b) 1116 (c) 2262 (d) 4495
Why: (a) 4249 ÷ 7 = 607. (b) 124 × 9 = 1116. (c) 5000 − 2738 = 2262. (d) 6400 − 1905 = 4495.

10
page 9

Answer: 24
Why: 6 × $3 = $18 and 4 × $2 = $8. The cost is $18 + $8 = $26. The change is $50 − $26 = $24.

11 page 9

Answer: (a) 13, 35 (b) 5, 17
Why: (a) 4 + 9 = 13. Then 9 + 13 = 22 and 13 + 22 = 35. (b) 11 + 6 = 17 and 6 + 11 = 17. The first number is 11 − 6 = 5. Check: 5 + 6 = 11, 6 + 11 = 17 and 11 + 17 = 28.

12 page 9

Answer: 240
Why: Aisha has 1 unit and Kumar has 3 units: 4 units in all. 1 unit = 480 ÷ 4 = 120. Kumar has 360 and Aisha has 120. The difference is 2 units: 240.

13 page 9

Answer: (a) Ali (b) 18
Why: Siti reads 24 × 15 = 360 pages. Ali reads 18 × 21 = 378 pages. Ali reads 378 − 360 = 18 more pages.

THINK HARDER

1 page 10

Answer: (a) 28 (b) 94
Why: (a) 4 × 9 = 36 and 12 ÷ 3 = 4. Then 60 − 36 + 4 = 24 + 4 = 28. (b) Brackets first: 13 + 7 = 20. Then 5 × 20 = 100 and 100 − 6 = 94.

2 page 10

Answer: 7
Why: The 4th number is 66 − 41 = 25. The 3rd is 41 − 25 = 16. The 2nd is 25 − 16 = 9. The 1st is 16 − 9 = 7. The chain is 7, 9, 16, 25, 41, 66.

3 page 10

Answer: (a) 18 (b) 90
Why: Mei is 5 units and Lily is 1 unit. After the gift they are equal, so Mei had 4 units more than Lily, and 4 units is 36 + 36 = 72. 1 unit = 18. Lily had 18 and Mei had 90. Check: 90 − 36 = 54 and 18 + 36 = 54.

GO FURTHER

1 page 11

Answer: (a) 9 × 4 − 6 (b) (7 + 3) × 8 − 4
Why: (a) 9 × 4 = 36 and 36 − 6 = 30. The other signs do not give 30. (b) (7 + 3) × 8 − 4 = 10 × 8 − 4 = 76. Brackets round 3 × 8 give 31, and brackets round 8 − 4 give 19.

2 page 11

Answer: 7
Why: Try 6: 6 × 6 + 6 = 42, too small. Try 8: 8 × 8 + 8 = 72, too big. Try 7: 7 × 7 + 7 = 49 + 7 = 56.

3 page 11

Answer: (a) 3 (b) 39 (c) 35
Why: Plan A has $15, $23, $31, $39. Plan B has $0, $13, $26, $39. Both have $39 in week 3. Each week Plan B gains $5 on Plan A, and the start gap is $15, so 15 ÷ 5 = 3 weeks. After 10 weeks Plan A has 15 + 80 = $95 and Plan B has $130, so Plan B has $35 more.

4 page 11

Answer: (a) 3 (b) 3
Why: 4 × 7 = 28, which ends in 8, as needed. 3□4 × 7 = 2128 + 70 × □ must be 2□38. Trying 3 gives 2128 + 210 = 2338. Check: 334 × 7 = 2338. So both missing digits are 3.

CHECK YOURSELF

1
page 12

Answer: 23 640 and 1305
Why: 4728 × 5 = 23 640 (4728 × 10 = 47 280, then half). 9135 ÷ 7 = 1305, because 7 × 1305 = 9135.

2
page 12

Answer: 116
Why: 15 × 12 = 180, so he saves $180. $180 – $64 = $116.

3
page 12

Answer: 220
Why: Blue is 1 unit and red is 4 units: 5 units. 1 unit = 275 ÷ 5 = 55. Red is 4 × 55 = 220.

4
page 12

Answer: 21
Why: Work backwards. 94 + 20 = 114. 114 ÷ 3 = 38. 38 – 17 = 21.